

Project Executable Files

Date	15 July 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes



Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
HDI-Predictor/
├── dataset/
│   └── Human Development Index and Components.csv
├── model/
│   └── hdi_model.pkl
├── static/
│   └── style.css
├── templates/
│   └── index.html
├── visualizations/
│   ├── correlation_heatmap.png
│   ├── scatter_plot.png
│   ├── strip_plot.png
│   └── correlation_matrix.csv
├── docs/
│   └── Project Documentation PDFs
├── app.py
├── train_model.py
├── visualize.py
├── requirements.txt
├── README.md
└── .gitignore
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server
Repository Link	https://github.com/Dimple1603/HDI-Predictor
Demo Video Link	https://drive.google.com/file/d/1MHFqWLGB4xlaiPU-kWKMbo_EH4xlo6qt/view?usp=drivesdk

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- Python 3.12 or above
- Visual Studio Code
- Git (Optional)
- Internet connection (for installing dependencies)

Step 1: Clone the repository

git clone <https://github.com/Dimple1603/HDI-Predictor.git>

Step 2: Open the project folder

cd HDI-Predictor

Step 3: Install the required dependencies

pip install -r requirements.txt

Step 4: Train the machine learning model

python train_model.py

Step 5: Start the Flask application

python app.py

Step 6: Open your browser and visit

<http://127.0.0.1:5000>

Enter the required values (Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita) and click "Predict HDI" to view the predicted Human Development Index score and development category.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application runs locally only	Can be deployed on Render later
2	Limited dataset size	Larger datasets can improve predictions
3	Linear Regression model only	Future work includes advanced ML models